

Matrix Model and Uberbau

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Figure 1: Matrix Model and Uberbau.

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1 Abstract

“Matrix Model and Uberbau” is a real-time simulation reflecting on the synchronicity between the evolution of computer graphics and AI technology and the turbulence of financial markets. The year 2008 stands as a stark mark, when financial and housing markets globally were ravaged by a devastating crisis following a period of deceptive prosperity. Amidst this economic turmoil, Nvidia unveiled CUDA, making general-purpose computing on the Graphics Processing Unit (GPU) streamlined and accessible. Traditionally

used to produce images rooted in military and entertainment, GPU has since become the essential apparatus for AI and infiltrated numerous domains including scientific computing and finance, due to its prowess in parallel computing and matrix operations. The artwork reflects on this history, with various scripted events driven by “predictions” of the 2007-08 Shanghai Stock Exchange Index using a deep learning neural network trained on the first-gen consumer GPU with CUDA support. In a space half sports stadium and half construction site, men and women in business suits running on the sports field are knocked over repeatedly by offscaled office items, spotlighted by GPU-drones equipped with searchlights. This scene of competition and violence is observed by automatic camera cuts and accompanied by the audio of Hong Kong comedian Wong Tze-wah’s gigs about financial and housing markets and an AI-cloned Zizek discussing graphics technology and ideology. The work asks: could the widespread adoption of AI, powered by computer graphics technology, provide us with the foresight into disasters previously unforeseen, or will it only further bolster the self-perpetuating hollow nature of finance capitalism? This question cascades into multiple subtopics—basic needs versus luxury; the notion of “model” in graphics, sciences, and economics; the emotional aspects of the markets; the uncertainty of life; and the paradox that the only perfect model might be the one that embraces its inherent inaccuracy.

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